

Periodic Geometry, Finite Attraction, and Exact Zeta Factors for Cyclic Rule 184

Route-A structural certificate C175

Abstract

For every ring length $N \geq 1$ and every conserved particle number $0 \leq k \leq N$, we classify the periodic states of cyclic Rule 184 and prove finite attraction to them. Below half density, the periodic core consists exactly of words without cyclic 11 and the rule is right rotation there; above half density, holes are isolated and the rule is left rotation. A gap-defect Lyapunov argument gives an explicit m^2 entry bound, where $m = \min(k, N - k)$. Cyclic independent-set counts then yield a closed formula for every iterate, all primitive temporal cycles, the Artin–Mazur zeta, and the periodic-core Koopman determinant. The whole sector is a permutation exactly when $m \leq 1$; otherwise its transients block full-system unitarity.

Keywords: Rule 184; traffic cellular automaton; periodic core; gap Lyapunov function; cyclic independent set; dynamical zeta.

中文摘要

本文对任意环长 N 和固定粒子数 k 完整刻画循环Rule 184的周期状态，并证明所有其他状态在有限时间内进入周期核。低密度时粒子最终彼此隔离并向右平移；高密度时空穴彼此隔离并向左平移。间隙缺陷的单调量给出 m^2 的进入上界，其中 $m = \min(k, N - k)$ 。由循环独立集计数进一步导出任意迭代的不动点闭式、本原周期、Artin–Mazur动力 ζ 函数及周期核Koopman行列式。含至少两个少数符号的全扇区存在瞬态，因而不能被宣称为酉系统。

关键词：Rule 184；交通元胞自动机；周期核；间隙Lyapunov函数；循环独立集；动力 ζ 函数。

1 Model and complete periodic core

For labelled cyclic sites, set

$$X_{N,k} = \left\{ x \in \{0,1\}^{\mathbb{Z}/N\mathbb{Z}} : \sum_i x_i = k \right\}, \quad F(x)_i = x_{i-1}(1 - x_i) + x_i x_{i+1}.$$

Thus every cyclic 10 is updated simultaneously to 01. One update is the clock, $(\rho x)_i = x_{i-1}$ is right rotation, and $m = \min(k, N - k)$.

Theorem 1 (All-size, all-density classification). *If $k \leq N/2$, a state is periodic exactly when it has no cyclic 11, and $F = \rho$ on that core. If $k \geq N/2$, a state is periodic exactly when it has no cyclic 00, and $F = \rho^{-1}$ on that core. Every state reaches the applicable core within m^2 updates. Every temporal least period divides N and equals the least cyclic-rotation period of the word.*

At half filling the two conditions leave exactly the two alternating states, both of period two. At $k = 0, N$, the unique uniform state is fixed. These statements include $N = 1$ and $N = 2$ without changing the update convention.

2 Gap defects and finite attraction

Assume first $0 < k \leq N/2$ and label the $m = k$ particles in cyclic order. Let $g_i \geq 0$ be the number of holes between particle i and its successor, and put $u_i = \mathbf{1}_{g_i > 0}$. Particle i moves precisely when $u_i = 1$; the simultaneous gap update is

$$g'_i = g_i - u_i + u_{i+1}. \tag{1}$$

A new zero at index i occurs exactly when the old $g_{i+1} = 0$ and $g_i \leq 1$. Hence every old zero-gap marker shifts one index backward unless its predecessor is at least two, in which case that marker is absorbed. The defect count $Z(g) = \#\{i : g_i = 0\}$ never increases.

Suppose $Z(g) > 0$. Since $\sum_i g_i = N - k \geq k = m$, some gap is at least two. An excess gap remains at its index until a zero arrives immediately after it: if its successor is positive, both terms in (1) cancel; if the successor is

zero, the excess absorbs that zero and decreases by one. Every unabsorbed zero moves backward once per update, so within at most m steps a zero meets an excess and Z strictly decreases. Initially $Z \leq m$; therefore after at most m^2 updates every gap is positive. This is exactly the no-11 core, where every particle advances and $F = \rho$.

Above half density, holes move left. Exchanging particles with holes and reversing cyclic orientation reproduces (1), proving attraction to the no-00 core and $F = \rho^{-1}$ there. Conversely each core word is periodic because rotation has order dividing N . If a periodic non-core word entered the forward-invariant core, a sufficiently large multiple of its period would put the original word in the core, a contradiction. This proves completeness rather than merely eventual regularity.

3 All iterates, primitive cycles, and zeta

Let $I(g, r)$ count cyclic length- g words with r isolated marked sites:

$$I(g, r) = \begin{cases} 1, & r = 0, \\ \frac{g}{g-r} \binom{g-r}{r}, & 1 \leq r \leq \lfloor g/2 \rfloor, \\ 0, & \text{otherwise.} \end{cases}$$

For $r > 0$, count linearly nonadjacent words and remove those with both endpoints marked, with out-of-range binomial coefficients interpreted as zero:

$$\binom{g-r+1}{r} - \binom{g-r-1}{r-2} = \frac{g}{g-r} \binom{g-r}{r}.$$

Theorem 2 (Every-iterate fixed count). *For $n \geq 1$, put $g = \gcd(N, n)$ and $q = N/g$. Then*

$$A_{N,k}(n) := \# \text{Fix}(F^n | X_{N,k}) = \begin{cases} I(g, m/q), & q \mid m, \\ 0, & q \nmid m. \end{cases}$$

An F^n -fixed word is periodic and hence lies in the isolated-minority core. There F^n is rotation by n in one of the two directions. Its invariant words are q repetitions of a cyclic block of length g . Divisibility $q \mid m$ is necessary, and the block contains m/q isolated minority symbols, proving the formula. At half filling it gives two fixed points exactly for even n ; at $k = 0, N$ it gives one for every n .

For $d \mid N$, Möbius inversion separates repeated from primitive data:

$$E_{N,k}(d) = \sum_{e \mid d} \mu(d/e) A_{N,k}(e), \quad P_{N,k}(d) = \frac{E_{N,k}(d)}{d}.$$

Here E counts labelled points of exact period d , while P counts geometric primitive cycles. Consequently

$$\zeta_{N,k}(z) = \prod_{d \mid N} (1 - z^d)^{-P_{N,k}(d)}, \quad \det(I - zU_{\text{core}}) = \prod_{d \mid N} (1 - z^d)^{P_{N,k}(d)}.$$

The determinant belongs to the finite permutation on the periodic core; it is exactly the reciprocal source zeta and is not a determinant invented for the transient full system.

4 Exact Koopman boundary

Every word in $X_{N,k}$ has isolated minority symbols exactly when $m \leq 1$. Thus the whole-sector map is a rotation permutation exactly in those sectors, and its uniform Koopman composition operator is unitary. If $m \geq 2$, a word with adjacent minority symbols exists and is transient by the classification theorem. A finite self-map with a transient state is not bijective, so its uniform whole-sector Koopman operator is not unitary.

The periodic core remains canonical for every sector. Restriction gives a finite rotation permutation, and spatial reflection reverses that rotation. This is a natural source-level operator on the core, but the restriction deletes transient traffic states. The all-sector family therefore receives only `A4_FORMAL_HINT`, not natural quantization of the original full phase spaces.

5 Evidence and Route-A decision

The finite ledger exhausts all 8,190 binary words through $N = 12$, all 90 sectors, 1,636 fixed-count rows through $n = 2N + 2$, 299 primitive rows, and 196,608 word-iterate comparisons. A producer-independent moving-particle

checker passes 34,545 assertions; SymPy passes 25,563 independent-set, Möbius, product, and gap checks; byte replay is exact; all 17 hostile mutations are rejected. These computations are regression sentinels. The all-parameter statements rest on the proofs above. Citation and reference registries each have population zero, and no external novelty or priority claim is made.

Gate	Verdict	Reason
A0	FAIL	no intrinsic arithmetic origin
A1	WEAK	complete intrinsic cycles, no arithmetic information
A2	FAIL	exact source zeta, no target divisor comparison
A3	FAIL	finite rational source, no target global structure
A4	FORMAL_HINT	core unitary discards transients for $m \geq 2$

The v0.2 tuple is (A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT), with overall ROUTE_A_REJECTED; Route B is false.

Limitations. The result concerns the frozen synchronous cyclic rule and labelled sectors. It does not establish asynchronous or open-boundary traffic laws. Exact finite zeta factors do not supply prime semantics. There is no target divisor, functional equation, counting law, continuation, Weil compression, arithmetic local factor, Euler factor, root number, automorphy claim, or Hilbert–Pólya operator. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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